

Perimeter and Area of Polygons on the Coordinate Plane

Answer Key

High School Geometry

Quick Answers

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|--|---|
| 1. 54 square units | perimeter = 20 units, (d) area = 15 square units |
| 2. (a) $AB = 10$ units, (b) $BC = 6$ units, (c) $CA = 8$ units, (d) perimeter = 24 units | 7. (a) area of the first plot = 30 square units, (b) area of the second plot = 36 square units, (c) difference = 6 square units |
| 3. 24 square units | 8. (a) area = 84 m^2 , (b) length of $CD = 7.21 \text{ m}$, — |
| 4. 72 m^2 | |
| 5. (a) area = 44 m^2 , — | |
| 6. (a) $WX = 5$ units, (b) $XY = 5$ units, (c) | |
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What is verified

6 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 5, 8. The worked solution states what a correct response must contain.

Curriculum

Level: High School Geometry
HSG-GPE.B.7 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 19 tagged checks.

Common wrong answers

- If they got 108: multiplied the two legs and forgot to halve*
 - If they got 25: multiplied two adjacent side lengths as if the figure were a rectangle*
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1. Area of the triangle $A(0, 0)$, $B(9, 0)$, $C(9, 12)$.

Solution. The legs $\overline{AB}$ and $\overline{BC}$ are horizontal and vertical, of lengths 9 and 12, so the triangle is right-angled at B :

$$\text{area} = \frac{1}{2}(9)(12) = \frac{1}{2}(108).$$

Therefore

54 square units

2. Triangle $A(-3, 2)$, $B(5, 8)$, $C(5, 2)$. (a) AB (b) BC (c) CA (d) perimeter.

Solution (a). $AB = \sqrt{(5 - (-3))^2 + (8 - 2)^2} = \sqrt{64 + 36} = \sqrt{100}$, so $AB = 10$ units

Solution (b). $\overline{BC}$ is vertical, so its length is the difference of the y -values: $8 - 2$, giving

$BC = 6$ units

Solution (c). $\overline{CA}$ is horizontal: $5 - (-3)$, giving

$CA = 8$ units

Solution (d). $10 + 6 + 8$, so

perimeter = 24 units

3. Area of the quadrilateral $(1, 1)$, $(7, 1)$, $(9, 5)$, $(3, 5)$.

Solution. Shoelace, going round in the given order:

$$(1 \cdot 1 - 7 \cdot 1) + (7 \cdot 5 - 9 \cdot 1) + (9 \cdot 5 - 3 \cdot 5) + (3 \cdot 1 - 1 \cdot 5) = -6 + 26 + 30 - 2 = 48,$$

and half of 48 is the area. Therefore

24 square units

Check: the figure is a parallelogram with base 6 and height 4.

4. Area of the park $(0, 0)$, $(10, 0)$, $(12, 6)$, $(2, 8)$, one metre per unit.

Solution. Shoelace:

$$(0 \cdot 0 - 10 \cdot 0) + (10 \cdot 6 - 12 \cdot 0) + (12 \cdot 8 - 2 \cdot 6) + (2 \cdot 0 - 0 \cdot 8) = 0 + 60 + 84 + 0 = 144.$$

Half of 144 gives

72 m²

5. L-shaped patio $(0, 0)$, $(8, 0)$, $(8, 4)$, $(4, 4)$, $(4, 7)$, $(0, 7)$, one metre per unit. (a) Area. (b) Check by splitting into two rectangles.

Solution (a). Shoelace round the six vertices:

$$0 + 32 + 16 + 12 + 28 + 0 = 88,$$

and half of 88 gives

44 m²

Solution (b) — **instructor-judged.** A complete response splits the patio into two rectangles — for example the 8×4 block along the bottom and the 4×3 block above its left half — computes 32 and 12, and adds them to 44, matching the shoelace result. Any valid decomposition with both partial areas and their sum earns full credit; naming the pieces without computing them does not.

6. Quadrilateral $W(-2, 1)$, $X(3, 1)$, $Y(7, 4)$, $Z(2, 4)$. (a) WX (b) XY (c) perimeter (d) area.

Solution (a). $\overline{WX}$ is horizontal: $3 - (-2)$, so

$WX = 5$ units

Solution (b). $XY = \sqrt{(7 - 3)^2 + (4 - 1)^2} = \sqrt{16 + 9} = \sqrt{25}$, so

$XY = 5$ units

Solution (c). All four sides measure 5, so the perimeter is 4×5 , giving

perimeter = 20 units

Solution (d). Shoelace:

$$(-2 - 3) + (12 - 7) + (28 - 8) + (2 + 8) = -5 + 5 + 20 + 10 = 30,$$

so the area is half of 30:

15 square units

Note: the figure is a rhombus, not a square — equal sides do not make the area 5×5 .

7. Two triangular plots: $(0, 0)$, $(10, 0)$, $(4, 6)$ and $(0, 0)$, $(8, 0)$, $(3, 9)$. (a) Area of the first. (b) Area of the second. (c) Difference.

Solution (a). Base 10 along the x -axis, height 6: $\frac{1}{2}(10)(6)$, so

30 square units

Solution (b). Base 8, height 9: $\frac{1}{2}(8)(9)$, so

36 square units

Solution (c). The second is larger: $36 - 30$, so

6 square units

8. Pentagon $A(0, 0)$, $B(12, 0)$, $C(12, 5)$, $D(6, 9)$, $E(0, 5)$, one metre per unit. (a) Area. (b) CD . (c) Why is the shoelace area independent of the starting vertex?

Solution (a). Shoelace:

$$0 + 60 + 78 + 30 + 0 = 168,$$

and half of 168 gives

84 m²

Solution (b). $CD = \sqrt{(6 - 12)^2 + (9 - 5)^2} = \sqrt{36 + 16} = \sqrt{52} = 7.2111\dots$, so

$CD = 7.21$ m

Solution (c) — instructor-judged. A complete response says the shoelace sum contains exactly one term per edge, and starting at a different vertex only rotates the order in which those same edge terms are added; addition does not depend on order, so the total — and therefore the area — is unchanged. A response that merely asserts the answer is the same, or that tries one other starting vertex without explaining why, earns partial credit.